1)

The tangential component of the electric field intensity is always continuous at the boundary. We get this by deriving that $E_{1t} = E_{2t}$ from the integral form of Maxwells equations and solving around the boundary of two different media. Drawing it out we can also clearly tell that for the electric field intensity to be continuous, the tangential component lying on the boundary has to be equal.

2)

We have the boundary condition for the normal components of D:

$$a_{n2} \cdot (\mathbf{D}_1 - \mathbf{D}_2) = \rho_s$$

So given $\sigma_f > 0$ the total surface charge density will be larger than 0, and thus the normal component of the dielectric will always be discontinuous at the boundary.

3)

The left hand side is the Poynting vector integrated over the closed volume of S, which equals the power leaving and entering the volume. This is referred to as Poynting's theorem.

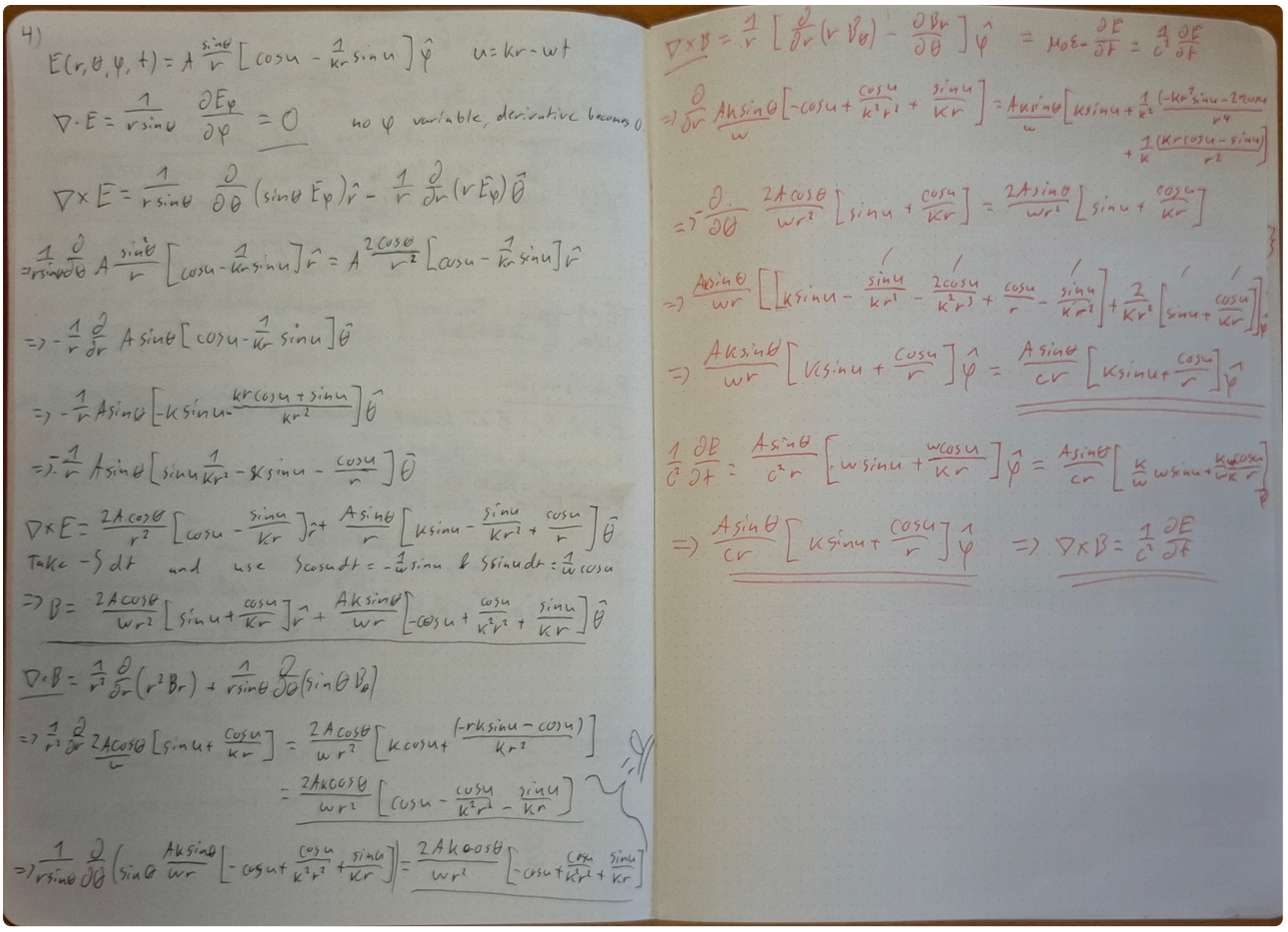
The first term on the right side gives the rate of change of energy stored in the electric and magnetic fields in the volume from time.

The second term on the right side gives the ohmic power dissipated in the volume due to the flow of conduction current density σE when in the presence of an electric field E. This is the source of heating due to Ohmic loss.

The last term on the right side gives the power given to the fields by an independant external source sending free currents. This could be a battery or an antenna that acts independant of the local E-field.

4)

a)


4)
$E(r,\theta,\varphi,t) = A\frac{\sin\theta}{r}\left[\cos u - \frac{1}{kr}\sin u\right]\hat{\varphi}$ $u = kr - \omega t$
$\nabla\cdot E = \frac{1}{r\sin\theta}\frac{\partial E_\varphi}{\partial\varphi} = 0$ no φ variable, derivative becomes 0.
$\nabla\times E = \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta E_\varphi)\hat{r} - \frac{1}{r}\frac{\partial}{\partial r}(rE_\varphi)\hat{\theta}$
$\Rightarrow \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}A\frac{\sin\theta}{r}\left[\cos u - \frac{1}{kr}\sin u\right]\hat{r} = A\frac{2\cos\theta}{r^2}\left[\cos u - \frac{1}{kr}\sin u\right]\hat{r}$
$\Rightarrow -\frac{1}{r}\frac{\partial}{\partial r}A\sin\theta\left[\cos u - \frac{1}{kr}\sin u\right]\hat{\theta}$
$\Rightarrow -\frac{1}{r}A\sin\theta\left[-k\sin u - \frac{kr\cos u + \sin u}{kr^2}\right]\hat{\theta}$
$\Rightarrow -\frac{1}{r}A\sin\theta\left[\sin u\frac{1}{kr^2} - k\sin u - \frac{\cos u}{r}\right]\hat{\theta}$
$\nabla\times E = \frac{2A\cos\theta}{r^2}\left[\cos u - \frac{\sin u}{kr}\right]\hat{r} + \frac{A\sin\theta}{r}\left[k\sin u - \frac{\sin u}{kr^2} + \frac{\cos u}{r}\right]\hat{\theta}$
Take $-\int dt$ and use $\int\cos u\,dt = -\frac{1}{\omega}\sin u$ & $\int\sin u\,dt = \frac{1}{\omega}\cos u$
$\Rightarrow B = \frac{2A\cos\theta}{\omega r^2}\left[\sin u + \frac{\cos u}{kr}\right]\hat{r} + \frac{Ak\sin\theta}{\omega r}\left[-\cos u + \frac{\cos u}{k^2r^2} + \frac{\sin u}{kr}\right]\hat{\theta}$
$\nabla\cdot B = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2B_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta B_\theta)$
$\Rightarrow \frac{1}{r^2}\frac{\partial}{\partial r}\frac{2A\cos\theta}{\omega}\left[\sin u + \frac{\cos u}{kr}\right] = \frac{2A\cos\theta}{\omega r^2}\left[k\cos u + \frac{(-rk\sin u - \cos u)}{kr^2}\right]$
$= \frac{2Ak\cos\theta}{\omega r^2}\left[\cos u - \frac{\cos u}{k^2r^2} - \frac{\sin u}{kr}\right]$
$\Rightarrow \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{Ak\sin\theta}{\omega r}\left[-\cos u + \frac{\cos u}{k^2r^2} + \frac{\sin u}{kr}\right]\right) = \frac{2Ak\cos\theta}{\omega r^2}\left[-\cos u + \frac{\cos u}{k^2r^2} + \frac{\sin u}{kr}\right]$
$\nabla\times B = \frac{1}{r}\left[\frac{\partial}{\partial r}(rB_\theta) - \frac{\partial B_r}{\partial\theta}\right]\hat{\varphi} = \mu_0\varepsilon_0\frac{\partial E}{\partial t} = \frac{1}{c^2}\frac{\partial E}{\partial t}$
$\Rightarrow \frac{\partial}{\partial r}\frac{Ak\sin\theta}{\omega}\left[-\cos u + \frac{\cos u}{k^2r^2} + \frac{\sin u}{kr}\right] = \frac{Ak\sin\theta}{\omega}\left[k\sin u + \frac{1}{k^2}\frac{(-kr^2\sin u - 2r\cos u)}{r^4} + \frac{1}{k}\frac{(kr\cos u - \sin u)}{r^2}\right]$
$\Rightarrow -\frac{\partial}{\partial\theta}\frac{2A\cos\theta}{\omega r^2}\left[\sin u + \frac{\cos u}{kr}\right] = \frac{2A\sin\theta}{\omega r^2}\left[\sin u + \frac{\cos u}{kr}\right]$
$\Rightarrow \frac{A\sin\theta}{\omega r}\left[k\sin u - \frac{\sin u}{kr^2} - \frac{2\cos u}{k^2r^3} + \frac{\cos u}{r} - \frac{\sin u}{kr^2}\right] + \frac{2}{kr^2}\left[\sin u + \frac{\cos u}{kr}\right]$
$\Rightarrow \frac{Ak\sin\theta}{\omega r}\left[k\sin u + \frac{\cos u}{r}\right]\hat{\varphi} = \frac{A\sin\theta}{cr}\left[k\sin u + \frac{\cos u}{r}\right]\hat{\varphi}$
$\frac{1}{c^2}\frac{\partial E}{\partial t} = \frac{A\sin\theta}{c^2r}\left[\omega\sin u + \frac{\omega\cos u}{kr}\right]\hat{\varphi} = \frac{A\sin\theta}{cr}\left[\frac{k}{\omega}\omega\sin u + \frac{k}{\omega}\frac{\omega\cos u}{kr}\right]\hat{\varphi}$
$\Rightarrow \frac{A\sin\theta}{cr}\left[k\sin u + \frac{\cos u}{r}\right]\hat{\varphi} \Rightarrow \nabla\times B = \frac{1}{c^2}\frac{\partial E}{\partial t}$

b)

4) b)

$$\vec{S} = \vec{E} \times \vec{H} = \frac{1}{\mu_0}(\vec{E} \times \vec{B}) = \frac{1}{\mu_0}\begin{pmatrix} E_r \\ E_\theta \\ E_\varphi \end{pmatrix} \times \begin{pmatrix} B_r \\ B_\theta \\ B_\varphi \end{pmatrix} = \frac{1}{\mu_0}\begin{pmatrix} 0 \\ 0 \\ E_\varphi \end{pmatrix} \times \begin{pmatrix} B_r \\ B_\theta \\ 0 \end{pmatrix}$$

$$= \frac{1}{\mu_0} E_\varphi \begin{pmatrix} -B_\theta \\ B_r \\ 0 \end{pmatrix} = \frac{1}{\mu_0} E_\varphi \left[-B_\theta \hat{r} + B_r \hat{\theta} \right]$$

$$\Rightarrow \frac{A \sin\theta}{\mu_0 r}\left[\cos u - \frac{\sin u}{kr}\right]\left\{\frac{2A\cos\theta}{\omega r^2}\left[\sin u + \frac{\cos u}{kr}\right]\hat{\theta} - \frac{Ak\sin\theta}{\omega r}\left[-\cos u + \frac{\cos u}{k^2 r^2} + \frac{\sin u}{kr}\right]\hat{r}\right\}$$

$$\Rightarrow \frac{A^2 \sin\theta}{\mu_0 \omega r^2}\left[\cos u - \frac{\sin u}{kr}\right]\left\{\frac{2\cos\theta}{r}\left[\sin u + \frac{\cos u}{kr}\right]\hat{\theta} + \sin\theta\left[k\cos u - \frac{\cos u}{kr^2} - \frac{\sin u}{r}\right]\hat{r}\right\}$$

$$\Rightarrow \vec{S} = \frac{A^2 \sin\theta}{\mu_0 \omega r^2}\left\{\frac{2\cos\theta}{r}\left[\cos u \sin u + \frac{\cos^2 u}{kr} - \frac{\sin^2 u}{kr} - \frac{\sin u \cos u}{k^2 r^2}\right]\hat{\theta} + \sin\theta\left[k\cos^2 u - \frac{\cos^2 u}{kr^2} - \frac{\cos u \sin u}{r} - \frac{\sin u \cos u}{r} + \frac{\sin u \cos u}{k^2 r^3} + \frac{\sin^2 u}{kr^2}\right]\hat{r}\right\}$$

$$\text{I} \Rightarrow \cos u \sin u = 0 \quad \text{and} \quad \cos^2 u = \sin^2 u = 1/2$$

$$\Rightarrow \frac{A^2 \sin\theta}{\mu_0 \omega r^2}\left\{\frac{2\cos\theta}{r}\left[0\right]\hat{\theta} + \sin\theta\left[\frac{k}{2}\right]\hat{r}\right\}$$

$$\vec{I} = \frac{A^2 \sin^2\theta}{2\mu_0 c r^2}\hat{r}$$ Points radially outwards and decreases by r^{-2}

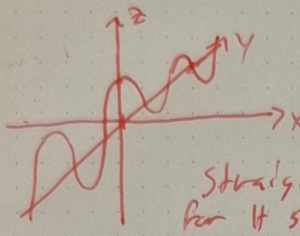
5 og 4 c)

c) $\int_0^{2\pi}\int_0^{\pi} I \cdot r^2 \sin\theta \, d\theta \, d\varphi = \frac{A^2}{2\mu_0 c}(2\pi)\int_0^{\pi}\sin^3\theta \, d\theta = \frac{A^2\pi}{\mu_0 c}\cdot\left(\frac{4}{3}\right)$

$\Rightarrow P = \frac{4\pi}{3}\frac{A^2}{\mu_0 c}$

5) $\tilde{E}(z) = \tilde{E}_0 \cdot e^{-jkz}$ $\quad E(z,t) = Re\{\tilde{E}(z)e^{j\omega t}\}$ $\quad E(0,t) = Re\{\tilde{E}_0 e^{j\omega t}\}$

a) $\tilde{E}(z) = \hat{y}\tilde{E}_0 e^{-jkz}$ $\quad E(0,t) = E_0\cos(\omega t)\hat{y}$ $\quad H(0,t) = \frac{1}{\eta}\hat{z}\times\tilde{E}_0$

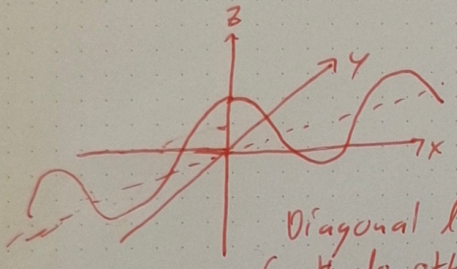


Straight line for H swap x and y

$H(0,t) = \frac{1}{\eta}E_0\cos(\omega t)\hat{z}\times\hat{y}$

$= \frac{1}{\eta}E_0\cos(\omega t)\hat{x}$

b) $\tilde{E}(z) = E_0 e^{-jkz}\hat{x} + E_0 e^{-jkz}\hat{y}$ $\quad E(0,t) = E_0\cos(\omega t)\hat{x} + E_0\cos(\omega t)\hat{y}$

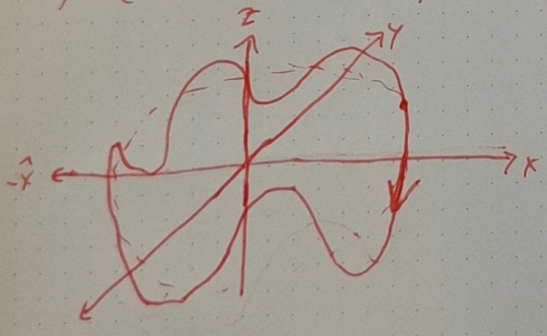


Diagonal line for H do other diagonal

$H(0,t) = \frac{1}{\eta}E_0\cos(\omega t)\begin{pmatrix}0\\0\\1\end{pmatrix}\times\begin{pmatrix}1\\1\\0\end{pmatrix} = \begin{matrix}-\hat{x}\\+\hat{y}\\0\end{matrix}$

$= \frac{1}{\eta}E_0\cos(\omega t)\hat{y} - \frac{1}{\eta}E_0\cos(\omega t)\hat{x}$

c) $\tilde{E}(z) = E_0 e^{-jkz}\hat{x} + jE_0 e^{-jkz}\hat{y}$ $\quad E(0,t) = E_0\cos(\omega t)\hat{x} - E_0\sin(\omega t)\hat{y}$



$H(0,t) = \frac{1}{\eta}E_0\begin{pmatrix}0\\0\\1\end{pmatrix}\times\begin{pmatrix}\cos(\omega t)\\-\sin(\omega t)\\0\end{pmatrix}$

$= \frac{1}{\eta}E_0\sin(\omega t)\hat{x} + \frac{1}{\eta}E_0\cos(\omega t)\hat{y}$

H is 90° out of phase.

I can't draw, but it's a circle going against the clock and up/down